Renminbi Exchange-Rate Expectations and China's Foreign Exchange Reserves: A Replication of Du, Ge and Qiao (2014)

Jiancheng Du

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Abstract

[WRITE LAST, 120-150 words: the original claim in one sentence; what we reproduce (cite the numbers of Section 4 with num()); the main errors found; whether the central conclusion survives the corrections, from Section 6.]

Keywords: replication, exchange-rate expectations, foreign exchange reserves, China, generated regressors, non-deliverable forwards.

JEL classification: F31, F32, E58, C22, C87.

Affiliation and contact: [AFFILIATION]; [EMAIL].

Conflict of interest: the author of this note is the first author of the study replicated here. The research was not funded by any party with an interest in its outcome. **[CONFIRM OR EDIT]**

1 Introduction

Du, Ge and Qiao (2014) estimate how expectations about the renminbi (RMB)–US dollar exchange rate affected the accumulation of China's foreign exchange reserves (FER) between 2000 and 2011. They split the sample into four exchange-rate regimes and measure expectations in two ways: a forecast generated from a three-stage regression model (Model I) and the 12-month non-deliverable forward rate (NDF, Model II). **[YOUR READING: in two or three sentences, the article's central claim as stated in its abstract and conclusion, quoted or closely paraphrased, with page numbers.]**

The question matters because **[YOUR READING: why this relation mattered, for example reserve accumulation and sterilisation costs, "hot money" inflows under an expected appreciation, and the policy debate at the time. One paragraph, with two or three references.]**

This note is a replication in the sense of the *Journal of Comments and Replications in Economics*: it assesses the reliability of the article's claims. We do two things. First, we **reproduce** the published estimates from the data of the original document on which the article is based, which contains the full dataset and every test and estimation table (Section 4). Second, we **correct** twelve problems in the original analysis, one at a time, and show how each correction changes the estimated effect of expectations (Sections 5 and 6).

Disclosure. The author of this note is the first author of the original article. To prevent the replication from being steered towards a preferred result, every rule that selects data, specifications or classifications was written down, dated, *before* the estimates it governs were computed, and the code and results of each step were committed to git. These rules include which balance-of-payments series is used, which unit-root specification, which Model I specification counts as the reproduction, and when an estimate counts as reproduced. Each rule is recorded, with the options considered and the evidence, in a numbered decision

log, which Appendix A lists. **[W1-09: the log is kept off the public repository; state whether a cleaned version is published with the package or is available on request.]** All code and all public data are available at **[REPOSITORY URL]**; the only inputs that cannot be redistributed are the original document and the Wind NDF series (Section 7). The co-authors of the original study were informed of this replication on **[DATE]**. **[CONFIRM OR EDIT]**

Section 2 describes the original study, Section 3 the data, Section 4 the reproduction, Section 5 the corrections, and Section 6 their effect on the conclusions. Section 7 explains how to reproduce every result, and Section 8 concludes.

2 The original study

2.1 Models

The article estimates two models of the monthly change in reserves, ΔFER_t, separately in four periods. **Model I** builds an expected change of the exchange rate in three stages. Stage 1 regresses the change of the exchange rate (RMB per USD) on fundamentals:

$$\Delta ER_t = b_0 + b_1\,\Delta BOP_t + b_2\,\Delta i_t + b_3\,\Delta \pi_t + b_4\,\Delta GDP_t + x_t. \tag{1}$$

Stage 2 projects the stage-1 result on its own lags, without a constant:

$$\widehat{\Delta ER}_t = \sum_{s=1}^{j} a_s\,\widehat{\Delta ER}_{t-s} + e_t, \tag{2}$$

and stage 3 uses the fitted value $\widehat{\widehat{\Delta ER}}_t$ as the expectation:

$$\Delta FER_t = \tau_0 + \tau_1\,\widehat{\widehat{\Delta ER}}_t + \tau_2\,\Delta NX_t + \tau_3\,\Delta FDB_t + \tau_4\,\Delta FDI_t + y_t. \tag{3}$$

Model II replaces the generated expectation by the observed 12-month NDF:

$$\Delta FER_t = g_0 + g_1\,\Delta NDF_t + g_2\,\Delta GDP_t + g_3\,\Delta NX_t + g_4\,\Delta FDB_t + g_5\,\Delta FDI_t + w_t. \tag{4}$$

The variables are defined in Table 2. In the original document, the regressions of Periods 1 and 3 have AR(1) errors (ARIMA(1,1,0)) and those of Periods 2 and 4 are estimated by OLS. The periods follow the exchange-rate regime (Table 1).

Table 1: Sample periods. Source: Du, Ge and Qiao (2014); original document, Chapter 3.

Period	From	To	Months
1	2000-01	2005-06	66
2	2005-07	2006-05	11
3	2006-06	2008-05	24
4	2008-06	2011-12	43

2.2 Reported findings

[YOUR READING: the article's Table 4 findings in the article's own words, one short paragraph: which expectation coefficients it reports as significant, with which sign, and how it interprets them. Page and table numbers.]

3 Data

3.1 The original data

All data and all reproduction targets are read by code from the original document; nothing is transcribed by hand. Appendix I of the original document supplies nine monthly variables for the 144 months from January 2000 to December 2011 (Table 2). The extracted data were checked against the original study's descriptive statistics (Tables 3.2 to 3.9): of the 304 cells of those tables, 267 agree to the printed decimals on the raw data (decision W2-13; tolerance: half a unit of the last printed digit, W2-07). **[YOUR READING: how the remaining cells are explained, from W2-13: for example the Period 1 correlations involving the constant exchange rate.]**

Table 2: Variables. Source: original Appendix I; SAFE; Wind; labels corrected where the original is inaccurate (Section 5).

symbol	label	source
FER	foreign exchange reserves	original Appendix I
ER	exchange rate, RMB per USD (a rise is a depreciation of the RMB)	original Appendix I
NDF	12-month USD/CNY non-deliverable forward, RMB per USD (Wind series in the source ledger)	Wind
BOP	balance of payments, current account (SAFE BPM6 row 1), quarterly converted to monthly (W2-10)	SAFE
GDP	GDP: cumulative year-to-date in the appendix; discrete quarterly in versions gdpq and corrected	original Appendix I
pi	CPI, same month of the previous year = 100 (a growth factor, not a price level)	original Appendix I
i	nominal one-year deposit rate (labelled 'real' in the article)	original Appendix I
NX	net exports	original Appendix I
FDI	foreign direct investment	original Appendix I
FDB	external debt stock	original Appendix I

3.2 Inputs not in the original appendix

Two regressors are not in the appendix and were collected anew, and price data were added for the corrections (Table 3; State Administration of Foreign Exchange (2026); Wind Information Co. (2026); International Monetary Fund (2026); Organisation for Economic Co-operation and Development (2026); National Bureau of Statistics of China (2026)).

- **Balance of payments.** The quarterly current account (SAFE, BPM6 presentation) is converted to months. Of three candidate conversions (quarterly value held, divided by three, year-to-date), the one used, bop_ca_ytd, was chosen by a rule fixed before any Model I estimate. The rule combines

the distance to the original unit-root statistic and to the four stage-1 Durbin–Watson statistics (W2-01, W2-10, W2-12).[1]

- **NDF.** The 12-month USD/CNY NDF is the only Wind tenor that covers 2000–2011. Month-end and month-average versions of the daily series were compared with the original unit-root statistic before Model II was estimated; the version used is ndf_mend (W2-06, W2-10).
- **Consumer prices.** For the deflator correction (Section 5.6), a fixed-base consumer price index for China was taken from the IMF data portal and cross-checked against FRED (OECD data). The NBS published year-on-year index (same month of the previous year = 100) was used to identify the original study's inflation series (W4-05, W4-14).

Table 3: Data sources. Coverage: first and last observation in the file as downloaded. Layer rep: used in the analysis; check: used only to verify another series. Source: data/metadata/source_ledger.csv of the replication package.

Series	Provider	Code	Frequency	Coverage	Method	Layer	Licence or access
Original document Appendix I (9 variables) and Tables 3.1-4.11	Original study document underlying Du, Ge & Qiao (2014)	App. I; T3.1, T3.2, T3.3, T3.4, T3.5, T3.6, T3.7, T3.8, T3.9, T4.1, T4.2, T4.3, T4.4, T4.5, T4.6, T4.7, T4.8, T4.9, T4.10, T4.11	monthly (App. I); period statistics and test tables	2000-01 to 2011-12	document extraction	rep	Unpublished original study document (co-author's); used for replication only. Not committed to git and not redistributed (decision W1-02).
BOP 1. 经常账户 (current account balance), BPM6, quarterly	State Administration of Foreign Exchange (SAFE)	sheet4:1. 经常账户	quarterly	1998-Q1 to 2026-Q2	file download	rep	Public official statistics (SAFE website); reproduction with attribution
无本金交割远期外汇 (NDF): 美元兑人民币:12 个月	Wind Information Co. (Wind Financial Terminal)	G0008386	daily	1999-01-19 to 2026-09-23	manual proprietary	rep	Proprietary Wind data exported by a licensed Wind user; not redistributable; kept in git-ignored data/raw/proprietary/
Consumer price index, all items, China (IMF CPI dataflow)	International Monetary Fund (IMF data portal)	IMF.STA/CPI/CHN.CPI._T.IX.M	monthly	1993-01 to 2026-08	api	rep	IMF data, free of charge; cite the IMF data portal (IMF terms of use)

[1] The original document predates SAFE's BPM6 presentation and probably used BPM5 data, so a residual mismatch may reflect the accounting standard rather than the method.

Series	Provider	Code	Frequency	Coverage	Method	Layer	Licence or access
Consumer Price Indices (CPIs, HICPs), COICOP 1999: Consumer Price Index: Total for China	FRED, Federal Reserve Bank of St. Louis (source: OECD Main Economic Indicators)	CHNCPIALLMINMEI	Monthly	1993-01 to 2025-04	api	check	OECD data via FRED; cite OECD and FRED as FRED requests
居民消费价格指数 (上年同月 =100) (CPI, same month of the previous year = 100)	National Bureau of Statistics of China (NBS), data.stats.gov.cn	nbs: 居民消费价格指数 (上年同月 =100)	monthly	1998-01 to 2015-12	file download	check	Public official statistics (NBS); cite the National Bureau of Statistics

4 Reproduction

4.1 Procedure

The reproduction follows rules fixed before estimation. Because the original document does not fully document Model I, all twelve combinations of its undocumented choices were estimated. These choices are whether stage 2 projects the stage-1 fitted values or the observed exchange rate, the number of lags (one to three), and whether the variables are used as published or divided by $\pi/100$, as the article states. The combination that reproduces most published cells was then named the headline (W3-02). Each estimate is classified on its sign and star level only (W1-07), with the meanings of the marks taken from the notes of the original tables and significance meaning $p < 0.10$. An estimate is **reproduced** if it has the same sign and the same star level as the published one, or if both are insignificant; **partially reproduced** if both are significant with the same sign but a different star level; and **not reproduced** if significance flips, or if the signs differ and either estimate is significant. Magnitude, measured as the gap in published standard errors, is reported but does not affect the class. First differences are taken on the whole 2000–2011 series (the "full" convention, W3-01); differencing within each period is reported as a sensitivity check.

4.2 Unit-root tests

The article reports augmented Dickey–Fuller (ADF) tests; the original tables report DF-GLS tests (Elliott et al. 1996). The original document does not state its lag rule. The specification that best reproduces the original statistics of the eight appendix variables was selected and then fixed (W2-02, W2-09). Table 4 shows the original statistics, our DF-GLS statistics and the ADF statistics at the same lag. DF-GLS and ADF agree on rejection at 5% in 14 of 14 series. **[YOUR READING: how close our DF-GLS statistics are to the original study's (compare the first two columns), and whether any stationarity judgement differs.]**

Table 4: Unit-root tests. Source: original Tables 4.1 and 4.2; own calculations.

variable	lags	original dfgls	dfgls	dfgls cv5	dfgls rejects	adf	adf cv5	adf rejects	original stationary
FER	1	0.249	0.249	-2.93	no	-1.861	-3.43	no	No

variable	lags	original dfgls	dfgls	dfgls cv5	dfgls rejects	adf	adf cv5	adf rejects	original stationary
GDP	1	-5.22	-5.221	-2.93	yes	-5.617	-3.43	yes	Yes
ER	1	-0.952	-0.93	-2.93	no	-1.881	-3.43	no	No
NX	1	-4.135	-4.135	-2.93	yes	-4.324	-3.43	yes	Yes
FDI	1	-6.72	-6.721	-2.93	yes	-7.712	-3.43	yes	Yes
FDB	1	-0.069	-0.069	-2.93	no	-0.344	-3.43	no	No
i	1	-1.805	-1.805	-2.93	no	-1.958	-3.43	no	No
DFER	1	-6.386	-6.386	-2.93	yes	-7.615	-3.43	yes	Yes
DGDP	1	-8.006	-8.006	-2.93	yes	-8.315	-3.43	yes	Yes
DER	1	-5.047	-5.01	-2.93	yes	-6.417	-3.43	yes	Yes
DNX	1	-8.543	-8.543	-2.93	yes	-8.905	-3.43	yes	Yes
DFDI	1	-14.724	-14.723	-2.93	yes	-18.242	-3.43	yes	Yes
DFDB	1	-11.026	-11.025	-2.93	yes	-11.556	-3.43	yes	Yes
Di	1	-5.748	-5.753	-2.93	yes	-5.984	-3.43	yes	Yes

[a] Original DF-GLS: original Tables 4.1 (levels) and 4.2 (first differences). Our DF-GLS and ADF use the specification fixed in Week 2 (deflated/trend/common/fixed_1) and the same number of lags.
[b] Critical values at 5%: DF-GLS from urca (Elliott, Rothenberg and Stock 1996 tables), ADF from urca (Dickey-Fuller tables). The original's critical values (Stata dfgls, finite-sample) differ by construction; compare the statistics.
[c] Rejects: the unit-root null is rejected at 5%.

4.3 Estimates

The headline Model I specification uses the alternative stage-2 regressor with 2 lag(s) on deflated data. It reproduces 19 of the 20 Model I cells; 0 are partially reproduced and 1 not reproduced. The ranking among the 12 combinations was decided by count (W3-07). For Model II on deflated data, 22 of 24 cells are reproduced (W3-08). Because two insignificant estimates count as reproduced whatever their size, the counts rest largely on such cells (W3-14): of the reproduced Model I cells, 14 are insignificant in both the original and our estimates, and 15 lie within one published standard error of the published value; for Model II, 18 and 20. Tables 5 and 6 (Tables 4.9 and 4.10 in the original document; Table 4 in the article) set the published and the reproduced estimates side by side, and Figure 1 shows each gap in units of the published standard error.

Table 5: Published and reproduced estimates, Periods 1 and 2. Source: original Table 4.9 (= article Table 4); own estimates.

	P1 I published	P1 I reproduced	P1 II published	P1 II reproduced	P2 I published	P2 I reproduced	P2 II published	P2 II reproduced
Expected DER (stage 3)	328.3027	-257.4538			164.2427	-722.7066		

	P1 I published	P1 I reproduced	P1 II published	P1 II reproduced	P2 I published	P2 I reproduced	P2 II published	P2 II reproduced
	(188.0208)	(1918.0945)			(1192.0540)	(1885.9070)		
DNDF			109.5668	111.4349			305.7626	117.0905
			(117.1090)	(116.9164)			(395.5048)	(331.0006)
DGDP			-0.0002	-0.0002			-0.0006	-0.0004
			(0.0003)	(0.0003)			(0.0006)	(0.0004)
DNX	-0.2176	-0.2424	-0.2420	-0.2369	0.5479	0.5163	0.3631	0.5091
	(0.4160)	(0.4696)	(0.5219)	(0.5200)	(0.7376)	(0.5833)	(0.6187)	(0.5670)
DFDB	-0.0900	-0.0336	-0.0264	-0.0250	-0.5976	-0.4209	0.1028	-0.3324
	(0.2196)	(0.1972)	(0.2393)	(0.2395)	(0.6362)	(0.5755)	(0.8068)	(0.5431)
DFDI	-0.3585	-0.2853	-0.1724	-0.1788	-0.9286	-0.2754	1.1845	-0.1552
	(0.5310)	(0.6569)	(0.6949)	(0.6944)	(2.3267)	(1.7917)	(2.8000)	(1.9957)
Constant	0.8782**	88.2383**	0.8737 **	87.3992**				
	(0.3081)	(29.5085)	(0.3388)	(33.8384)				
AR(1)	0.6565***	0.6525***	0.6663***	0.6659***				
	(0.0641)	(0.0729)	(0.0640)	(0.0642)				
N	65	63	65	65		11		11
Estimator	ARIMA	ARIMA	ARIMA	ARIMA	OLS	OLS	OLS	OLS

[a] Published: original Table 4.9 as printed (marks as printed); the numbers in parentheses are those printed below each coefficient, which the article labels t-statistics (see the section 'Standard errors labelled as t-statistics').

[b] Reproduced: our estimates, Model I with the alternative stage-2 regressor and 2 lag(s) on deflated data (rank 1 under the pre-specified rule W3-02; result W3-07); Model II on the same data; differences on the full convention. Standard errors in parentheses: OPG for ARIMA periods, OLS otherwise.

[c] n.i. = not identified: the regressor has no variation in that period (the exchange rate was pegged in Period 1).

[d] Estimator: ARIMA = regression in first differences with AR(1) errors (ARIMA(1,1,0) in levels), exact maximum likelihood; OLS otherwise.

[e] Our coefficients are rounded to the decimals printed in the original.

[f] Shading of the reproduced estimates (classification W1-07, on sign and star level only; significant = $p < 0.10$): green = reproduced (same sign and same star level, or both insignificant); yellow = partially reproduced (same sign, both significant, different star level); red = not reproduced (significance flips, signs differ with either significant, or not identified).

[g] Significance (legend of original Tables 4.9-4.10): † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-sided; normal for the ARIMA periods, t for OLS).

[h] Periods: 1 = 2000-01 to 2005-06; 2 = 2005-07 to 2006-05; 3 = 2006-06 to 2008-05; 4 = 2008-06 to 2011-12 (Replication Plan s.3.3).

Table 6: Published and reproduced estimates, Periods 3 and 4. Source: original Table 4.10 (= article Table 4); own estimates.

	P3 I published	P3 I reproduced	P3 II published	P3 II reproduced	P4 I published	P4 I reproduced	P4 II published	P4 II reproduced
Expected DER (stage 3)	-200.0742	462.5466			-4308.4840	386.9869		
	(476.6059)	(1016.0453)			(8324.3310)	(2575.8167)		
DNDF			-163.1845	-158.1010			-1294.4460*	-1282.9913*
			(376.3561)	(318.0285)			(531.2001)	(523.6424)
DGDP			-.0007	-0.0007			0.0008	0.0008
			(0.0007)	(0.0006)			(0.0007)	(0.0007)
DNX	0.2384	0.1956	.4362	0.4192	-0.5519	-0.5259	-0.8213	-0.8222
	(0.4853)	(0.4848)	(0.4542)	(0.4220)	(0.6736)	(0.6558)	(0.6352)	(0.6272)
DFDB	0.4041 **	0.4116**	.4403*	0.4542**	0.5843	0.6607†	0.4315	0.4312
	(0.1425)	(0.1594)	(0.1738)	(0.1596)	(0.3504)	(0.3506)	(0.3126)	(0.3087)
DFDI	-0.9285	-0.9432	-1.6370	-1.6336	-2.6346	-2.3344	-3.5087	-3.5591
	(0.9880)	(0.8122)	(1.3808)	(1.2962)	(2.8110)	(2.7688)	(2.5477)	(2.5060)
Constant	2.9458***	321.0177***	2.9920 ***	292.9954***				
	(0.6852)	(80.5416)	(0.7057)	(65.0397)				
AR(1)	0.5950 **	0.6549**	0.6647†	0.6821*				
	(0.2175)	(0.2146)	(0.3530)	(0.3243)				
N	23	24	23	24		43		43
Estimator	ARIMA	ARIMA	ARIMA	ARIMA	OLS	OLS	OLS	OLS

[a] Published: original Table 4.10 as printed (marks as printed); the numbers in parentheses are those printed below each coefficient, which the article labels t-statistics (see the section 'Standard errors labelled as t-statistics').
[b] Reproduced: our estimates, Model I with the alternative stage-2 regressor and 2 lag(s) on deflated data (rank 1 under the pre-specified rule W3-02; result W3-07); Model II on the same data; differences on the full convention. Standard errors in parentheses: OPG for ARIMA periods, OLS otherwise.
[c] n.i. = not identified: the regressor has no variation in that period (the exchange rate was pegged in Period 1).
[d] Estimator: ARIMA = regression in first differences with AR(1) errors (ARIMA(1,1,0) in levels), exact maximum likelihood; OLS otherwise.
[e] Our coefficients are rounded to the decimals printed in the original.
[f] Shading of the reproduced estimates (classification W1-07, on sign and star level only; significant = $p < 0.10$): green = reproduced (same sign and same star level, or both insignificant); yellow = partially reproduced (same sign, both significant, different star level); red = not reproduced (significance flips, signs differ with either significant, or not identified).
[g] Significance (legend of original Tables 4.9-4.10): † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

(two-sided; normal for the ARIMA periods, t for OLS).
[h] Periods: 1 = 2000-01 to 2005-06; 2 = 2005-07 to 2006-05; 3 = 2006-06 to 2008-05; 4 = 2008-06 to 2011-12 (Replication Plan s.3.3).

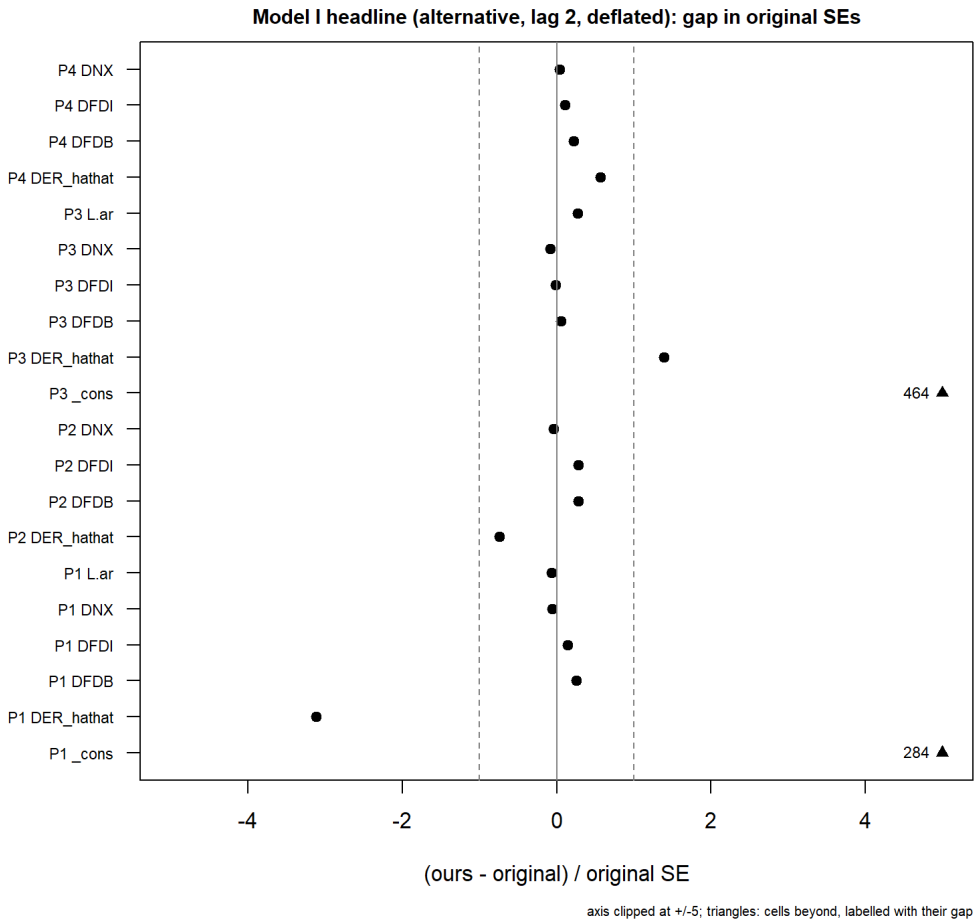


Figure 1: Gap between reproduced and published Model I estimates, in published standard errors (headline specification). Dashed lines: one standard error. Cells beyond the clipped axis are marked with triangles and labelled with their gap. Source: own estimates; original Tables 4.9-4.10.

[YOUR READING, from Table 5, Table 6 and W3-07: which terms reproduce (the controls, the AR terms) and which do not (the expectation term and the constants of Periods 1 and 3). For the constants, W3-07 records the ratio ours/original; say whether you read it as a difference in units or scaling of the constant, and mark it as your interpretation.]

The number of observations in the original document is matched in 2 of 28 comparisons (W3-11). **[YOUR READING: whether the original study appears to have differenced within periods, with reference to W3-11.]**

Table 7: Reproduction status by model and data version (primary convention). Model I: headline specification. Source: own classification (rules in W1-07).

model	version	n cells	reproduced	reproduced both insignificant	reproduced within one se	partially	not reproduced	mean std gap
I	deflated	20	19	14	15	0	1	37.77

model	version	n cells	reproduced	reproduced both insignificant	reproduced within one se	partially	not reproduced	mean std gap
I	raw	20	14	12	14	2	4	32.10
II	deflated	24	22	18	20	2	0	27.87
II	raw	24	20	17	15	2	2	29.66

4.4 Autocorrelation tests

The original study tests residual autocorrelation with the Durbin–Watson statistic and, where it is inconclusive, with the Breusch–Godfrey test (original Tables 4.3 to 4.8). Over all tables, data versions and conventions, 1 of 168 statistics agree with the printed values to the printed decimals, and 124 of 168 test decisions agree (W3-12). On deflated data under the primary convention, the decisions agree in 37 of 42 cases. **[YOUR READING: what you conclude from statistics that do not reproduce while most decisions do.]**

4.5 Chow tests

The original study reports Chow tests of equal coefficients in adjacent periods (original Table 4.11). Table 8 shows them next to three recalculations. In every model and pair, the published F statistic exceeds the largest recalculated one by a factor of at least $10^{-7.8}$ (W3-13). The published degrees of freedom match those of the same regressions estimated in levels, rather than in differences, in 11 of 12 comparisons. **[YOUR READING: what the degrees-of-freedom match suggests about how the published statistics were computed. Mark it as your interpretation.]**

Table 8: Chow tests between adjacent periods. Source: original Table 4.11; own calculations.

model	periods	published	classic differences	classic levels	robust wald
I	1-2	F (5, 67) =17677.62	F(5,64) = 2.79	F(5,67) = 0.12	F(5,121) = 7.90, p = 0.0000
I	2-3	F (5, 25) = 5736.75	F(5,25) = 0.41	F(5,25) = 5.20	F(5,121) = 6.58, p = 0.0000
I	3-4	F (5, 57) = 1.6e+06	F(5,57) = 0.12	F(5,57) = 41.47	F(5,121) = 1.37, p = 0.2399
II	1-2	F (6, 65) =16379.22	F(6,64) = 2.37	F(6,65) = 2.05	F(6,119) = 7.00, p = 0.0000
II	2-3	F (6, 23) = 7859.14	F(6,23) = 1.48	F(6,23) = 2.37	F(6,119) = 6.82, p = 0.0000
II	3-4	F (6, 55) =94985.79	F(6,55) = 1.08	F(6,55) = 45.56	F(6,119) = 2.79, p = 0.0141

[a] Published: original Table 4.11 as printed.
[b] Classic: Chow F with a common error variance, on the model as estimated in first differences, and on the same regressors in levels (whose degrees of freedom are compared with the published ones in the text).
[c] Robust Wald: equality of all coefficients of the two regimes in the fully interacted pooled model (Table 4), Newey-West covariance, F = W/q.

5 Corrections

We take the twelve problems listed in the replication plan one at a time. For each, we state what the original did, why that is a problem, what we do instead, and what changes. The data and model changes are cumulative, and we assess their combined effect in Section 6.

5.1 Standard errors labelled as t-statistics

The article describes the numbers in parentheses below each coefficient as t-statistics. The significance marks printed in the same table allow a direct test. If the parentheses held t-statistics, the marks would follow from those numbers; if they held standard errors, the marks would follow from the ratio of coefficient to parenthesis. Read as standard errors, the parentheses reproduce the printed marks in 42 of 44 cells (95%); read as t-statistics, in 22 (50%). The cells where the standard-error reading does not reproduce the marks are: Model I, Period 1, DER_hathat; Model I, Period 4, DFDB. **[YOUR READING: what the numbers in parentheses are, and whether the non-matching cells lie near a significance threshold (corr_relabel.csv, column implied_t_if_se).]** Throughout this note, the parentheses of the published tables are treated as standard errors.

5.2 Units: level coefficients and log differences

All variables of 1 to 4 are first differences of *levels*, so each coefficient is measured in the units of the variables in Table 2, for example reserves per unit change of the exchange rate. **[YOUR READING: quote the article's percentage interpretation of a coefficient, with page, and state why a level coefficient cannot be read as a percentage.]** To give the coefficients a unit-free reading, all models were re-estimated with log differences of the strictly positive series (FER, the exchange rate, the NDF, GDP, FDB and FDI). The expectation coefficient then is an elasticity; net exports, the balance of payments, inflation and the interest rate stay in level differences (row 6 of Tables 13 and 14).

5.3 Sign convention

Both expectation measures are quoted in RMB per US dollar, so a rise is an expected *depreciation* of the RMB. A positive coefficient therefore means that reserves grow faster when a depreciation is expected, and a negative one that they grow faster when an appreciation is expected. **YOUR READING: how the article describes the sign in its abstract and conclusion, and whether that description matches this convention. Use the Period 4 Model II coefficient: published sign from 6**

5.4 Generated regressors

In Model I, the stage-3 regressor is estimated in stages 1 and 2. The conventional stage-3 standard errors treat it as data and so ignore the sampling variation of the earlier stages (Pagan 1984; Murphy and Topel 1985). A bootstrap of the whole three-stage chain includes that variation. Stages 1 to 3 are re-estimated in each of 999 replications, with seed 20261001 (W4-01). The primary scheme is a wild bootstrap (Mammen 1993; Davidson and Flachaire 2008): the residuals of stage 1 and the innovations of stage 3 are multiplied by the same Rademacher draw in each month, which preserves their correlation, and the AR(1) errors of Periods 1 and 3 are rebuilt from the sign-flipped innovations. A moving-block bootstrap within periods (Künsch 1989) is reported as a check. Every coefficient was estimated in at least 999 of 999 replications.

For the headline specification on deflated data, the ratio of the bootstrap to the conventional standard error of the expectation coefficient is, by period: Period 1 0.55, Period 2 1.18, Period 3 1.02, Period 4 0.79 (wild) and Period 1 1.65, Period 2 2.03, Period 3 1.10, Period 4 2.35 (block). Figure 2 compares the intervals, and Figure 3 shows the bootstrap distributions. **YOUR READING: whether the conventional standard errors understate or overstate the uncertainty, and whether any significance mark changes (row 2 of 13**

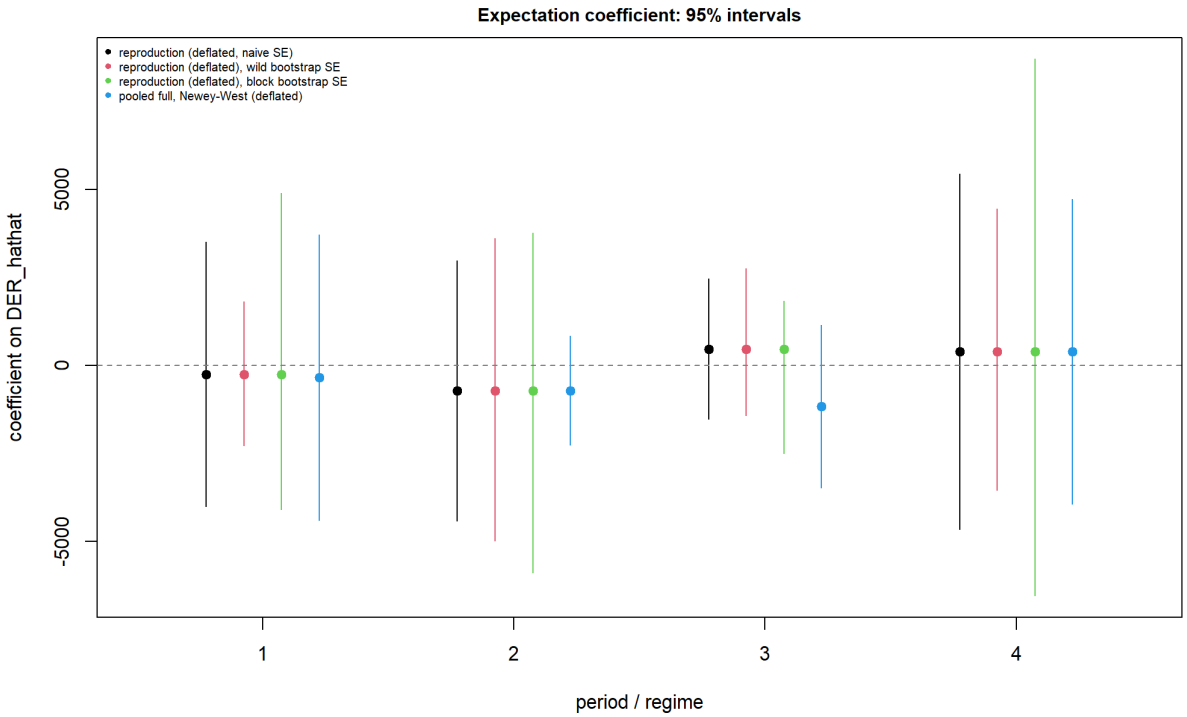


Figure 2: Expectation coefficient of Model I by period: 95% intervals from the conventional standard error, the wild and the block bootstrap (percentile), and the pooled model with Newey-West standard errors. Headline specification. Source: own estimates.

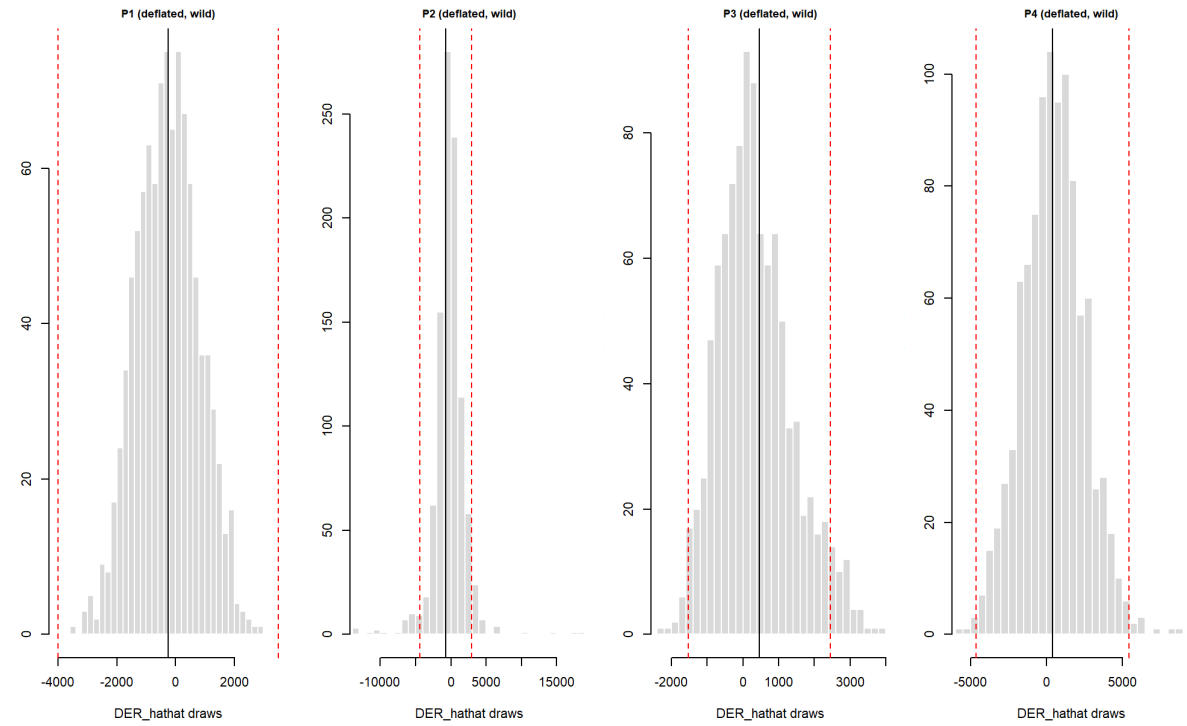


Figure 3: Wild-bootstrap distribution of the Model I expectation coefficient by period. Solid line: point estimate; dashed lines: conventional 95% interval. Source: own estimates.

5.5 Year-to-date GDP

In the appendix, the GDP of every month of a quarter is the cumulative GDP of the year to date. Its first difference is therefore zero in two months of each quarter, positive at each quarter start, and strongly negative every January, when the cumulation restarts. This January fall is an accounting artefact that enters both models as a regressor. The correction replaces it by discrete quarterly GDP (the difference of consecutive year-to-date values), held in each month of the quarter (W4-05). The code first verifies that the appendix value is constant within every quarter. The effect is shown in row 3 of Table 13 and Table 14, to be compared with row 1b, the reproduction on the same (raw) data.

5.6 Deflator

The article divides every variable by $\pi/100$. Here π is the consumer price index with the same month of the previous year = 100, a year-on-year growth factor P_t/P_{t-12} rather than a price level. Dividing by it gives $X_t P_{t-12}/P_t$: the real value multiplied by last year's price level, not a real series. The correction divides the value variables (FER, GDP, net exports, FDI, external debt and the balance of payments) by a fixed-base consumer price index, rebased to 2000 = 100, from the IMF. The exchange rate and the NDF are not deflated: they are prices of one currency in another, and a real exchange rate would require both countries' price levels (W4-05).[2]

We check the price data in two ways (Table 9 and Figure 4). First, we compare the original π with the same-month year-on-year index published by the National Bureau of Statistics (NBS): 144 of 144 months agree within the printed decimals, with a largest gap of 0.000 points (W4-14). **[YOUR READING: what this comparison shows about the source of the original π.]** Second, the year-on-year rates implied by the IMF index agree with the published ones within the printed decimals in 31 of 144 months (largest gap 0.592 points).[3] **[YOUR READING: from Figure 4, where the gaps between the published and the index-implied rates lie in time. "Our interpretation suggests …" (for example the NBS basket changes; mark as a hypothesis).]**

Table 9: Consumer-price checks, 2000-2011. Within print: gap no larger than half a unit of the last decimal printed in the original document. Source: original Appendix I; IMF; FRED; NBS; own calculations.

check	n	mean abs gap	max abs gap	n within print	correlation
original pi vs IMF CPI year-on-year	144	0.0764	0.5922	31	0.9989
IMF vs FRED CPI year-on-year (sample months)	144	0.0000	0.0000		1.0000
IMF / FRED index ratio (constant if same index, other base)	144	0.0000	0.0000		
original pi vs NBS published year-on-year	144	0	0	144	1
NBS published year-on-year vs IMF CPI year-on-year	144	0.0764	0.5922	31	0.9989

[2] The US-dollar variables are thereby deflated by China's consumer prices, which measures their purchasing power over Chinese goods.

[3] We compared the FRED series with the IMF's in the same way (largest year-on-year gap 0.0000). **[YOUR READING: whether FRED is an independent measure or the same index, and therefore what it confirms.]**

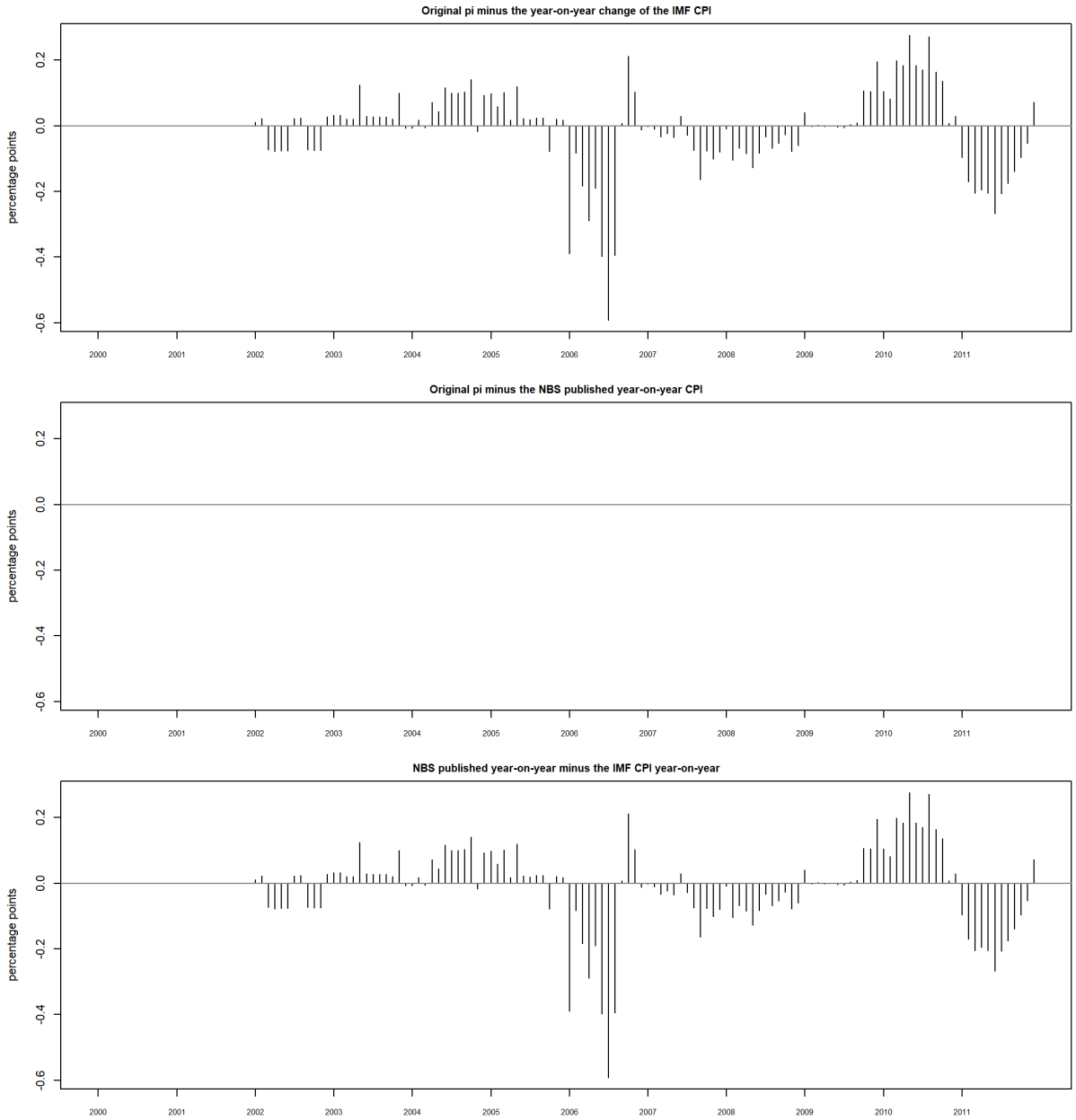


Figure 4: Gaps between the original inflation series, the NBS published year-on-year CPI and the year-on-year rate implied by the IMF CPI index, 2000-2011. Source: original Appendix I; NBS; IMF; own calculations.

5.7 Period 1: exchange-rate variation

From January 2000 to June 2005 the RMB was pegged to the US dollar, and the appendix exchange rate is constant (standard deviation 0.000 on raw data). Its change is zero, so no expectation effect can be estimated from it: the coefficient is not identified (W3-10). In the article's deflated data the exchange rate varies (standard deviation 0.138), and that variation comes from the deflator: the R^2 of the deflated exchange-rate change on the change of π is 0.999 (Table 10). A Period 1 "expectation effect" estimated on deflated data is therefore an effect of inflation. With the exchange rate undeflated (the corrected data), the Period 1 coefficient is reported as not identified.

Table 10: Period 1: variation of the exchange rate, its change, and the stage-3 expectation term, by data version. Source: own calculations.

version	sd ER level	sd DER	sd DER hathat	r2 DER on Dpi
raw	0	0	0	
deflated	0.1379	0.0530	0.0049	0.9991
gdpq	0	0	0	
corrected	0	0	0	

5.8 Unit-root test label

The article reports ADF tests; the original study ran DF-GLS tests. Table 4 reports both at the same lag, and they agree on rejection in 14 of 14 series. This correction relabels the tests and changes no conclusion that depends on them. **CONFIRM from 4**

5.9 Chow tests, correctly specified

The classic Chow test (Chow 1960) assumes equal error variances in the two regimes and no autocorrelation. Both assumptions are doubtful with monthly data across exchange-rate regimes. The corrected test estimates one regression over 2000–2011 in which every coefficient is regime-specific (Section 5.10), and tests the equality of all coefficients of two adjacent regimes with a Wald test based on Newey–West standard errors (last column of Table 8; Table 11). For the corrected data, Model I, the tests give: regimes 1–2, $F(4, 122) = 7.84$, $p < 0.001$; regimes 2–3, $F(5, 122) = 4.52$, $p < 0.001$; regimes 3–4, $F(5, 122) = 0.80$, $p = 0.553$. For Model II, in the same order: $F(6, 119) = 6.50$, $p < 0.001$; $F(6, 119) = 12.91$, $p < 0.001$; $F(6, 119) = 3.97$, $p = 0.001$. **[YOUR READING: whether the regime split of the original is supported, pair by pair.]**

5.10 Period 2: pooled model with regime interactions

Period 2 has about a year of monthly observations and five or six regressors, leaving very few residual degrees of freedom. Following the replication plan, the four periods are pooled in one regression with regime-specific constants. In the *full* variant every coefficient is regime-specific, which gives the same point estimates as the separate regressions but one set of standard errors for the whole sample. In the *expectation-only* variant only the constant and the expectation term are regime-specific and the controls are common (W4-02). Standard errors are Newey–West, with a lag of 4 (Newey and West 1987, 1994) (W4-03).

The expectation-only variant is valid only if the controls have the same effect in every regime. The Wald test of that restriction in the full model gives $F(9, 122) = 3.76$, $p < 0.001$ for Model I and $F(12, 119) = 3.55$, $p < 0.001$ for Model II (corrected data). The test that the expectation coefficient is the same in all regimes gives $F(2, 122) = 3.33$, $p = 0.039$ (Model I) and $F(3, 119) = 1.55$, $p = 0.206$ (Model II). **[YOUR READING: whether the equal-controls restriction is rejected, and therefore which of the two pooled variants is the appropriate one; then what the pooled model says about Period 2.]**

Table 11: Pooled models, 2000-2011, with regime interactions and Newey-West standard errors; corrected data. Source: own estimates.

	I full	I exp only	II full	II exp only
Expectation term, regime 1			-108.413	-116.098
			(222.184)	(223.077)
Expectation term, regime 2	94.719	-918.965	-18.652	551.521
	(584.259)	(850.136)	(160.708)	(429.832)
Expectation term, regime 3	-2371.261**	-2288.598**	130.540	72.686
	(850.386)	(864.105)	(344.420)	(278.563)
Expectation term, regime 4	1476.861	1556.231	-1506.262*	-1609.144*
	(3012.285)	(2891.659)	(741.241)	(712.074)
DGDP (common)				0.003
				(0.002)
DNX (common)		0.404		0.438
		(0.405)		(0.368)
DFDB (common)		0.643***		0.506***
		(0.138)		(0.135)
DFDI (common)		-0.159		-0.570
		(0.767)		(0.878)
All coefficients equal, regimes 1-2	F(4, 122) = 7.84, p < 0.001		F(6, 119) = 6.50, p < 0.001	
All coefficients equal, regimes 2-3	F(5, 122) = 4.52, p < 0.001		F(6, 119) = 12.91, p < 0.001	
All coefficients equal, regimes 3-4	F(5, 122) = 0.80, p = 0.553		F(6, 119) = 3.97, p = 0.001	
Controls equal in all regimes	F(9, 122) = 3.76, p < 0.001		F(12, 119) = 3.55, p < 0.001	
Expectation term equal in all regimes	F(2, 122) = 3.33, p = 0.039	F(2, 131) = 1.36, p = 0.260	F(3, 119) = 1.55, p = 0.206	F(3, 131) = 2.64, p = 0.052
N (Newey-West lag)	141 (4)	141 (4)	143 (4)	143 (4)

[a] Pooled OLS over 2000-2011, data version 'corrected', level form, with regime-specific constants. full: every coefficient regime-specific (point estimates equal the separate per-period OLS fits; controls not shown); exp_only: only the constant and the expectation term regime-specific, controls common.
[b] Newey-West standard errors in parentheses: Bartlett kernel, lag floor(4(n/100)^(2/9)), no prewhitening, small-sample factor n/(n-k); p-values from t(n-k). Results with the automatic bandwidth are in corr_pooled_coefs.csv and corr_pooled_tests.csv.
[c] Tests: Wald tests with the Newey-West covariance, reported as F = W/q with (q, n-k) degrees of free-

dom. A term not identified in a regime is left out of the restrictions.
[d] Model I uses the stage-3 expected change of the exchange rate from the headline specification; its pooled standard errors do not account for the generated regressor (see Table 3 for the bootstrap).
[e] Significance (legend of original Tables 4.9-4.10): † p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001 (two-sided; normal for the ARIMA periods, t for OLS).
[f] Periods: 1 = 2000-01 to 2005-06; 2 = 2005-07 to 2006-05; 3 = 2006-06 to 2008-05; 4 = 2008-06 to 2011-12 (Replication Plan s.3.3).

5.11 Interest-rate label

The article calls the interest rate "real". The appendix series is the nominal one-year deposit rate; Table 2 uses the correct label. The label does not affect any estimate.

5.12 NDF source

The article does not state the source of its NDF series. We use the 12-month USD/CNY NDF from Wind, identified by its Wind code in Table 3 (W2-06).[4] **[CONFIRM against W1-03 and W2-06.]**

5.13 Corrected estimates

Table 12 reports both models on the corrected data (discrete quarterly GDP, fixed-base CPI, exchange rates undeflated) with the original study's estimators. For Model I it also shows the wild-bootstrap standard error of the expectation term.

Table 12: Corrected estimates. Source: own estimates.

	P1 I	P1 II	P2 I	P2 II	P3 I	P3 II	P4 I	P4 II
Expected DER (stage 3)	n.i.		94.719		-2871.442		1476.861	
			(1152.605)		(2699.493)		(3395.443)	
			(1409.991)		(1558.607)		(5443.930)	
DNDF		8.040		-18.652		-377.679		-1506.262**
		(119.602)		(426.227)		(309.467)		(482.755)
DGDP		0.001		-0.001		-0.007*		0.006
		(0.003)		(0.005)		(0.003)		(0.005)
DNX	-0.147	-0.199	2.007*	1.974*	0.050	0.437	0.282	0.298
	(0.419)	(0.487)	(0.570)	(0.644)	(0.246)	(0.315)	(0.686)	(0.636)
DFDB	-0.083	-0.097	-0.855	-0.721	0.571**	0.699***	0.781*	0.572†
	(0.210)	(0.232)	(0.518)	(0.743)	(0.189)	(0.162)	(0.322)	(0.284)
DFDI	-0.117	-0.126	-1.133	-0.837	0.192	-0.374	0.027	-0.857
	(0.591)	(0.672)	(1.525)	(2.212)	(0.986)	(1.138)	(2.663)	(2.304)
Constant	85.026***	82.372***	177.225***	174.450***	132.224	238.069*	203.118**	180.647***

[4] Wind lists the series as compiled from news reports; its 1-, 3- and 6-month tenors start only at the end of 2008, so the 12-month tenor is the only one covering the sample.

	P1 I	P1 II	P2 I	P2 II	P3 I	P3 II	P4 I	P4 II
	(22.895)	(23.495)	(22.752)	(20.290)	(111.832)	(101.544)	(67.340)	(44.384)
AR(1)	0.569***	0.581***			0.542*	0.730†		
	(0.075)	(0.073)			(0.222)	(0.373)		
N	63	65	11	11	24	24	43	43

[a] Data version 'corrected': discrete quarterly GDP; FER, GDP, NX, FDI, FDB and BOP divided by the IMF consumer price index (2000 = 100); exchange rates, inflation and the interest rate not deflated.
[b] Model I: alternative stage-2 regressor, 2 lag(s) (the headline specification); same estimators as the original (ARIMA(1,0,0) by ML in Periods 1 and 3, OLS in Periods 2 and 4).
[c] Second line: conventional standard error (OPG in the ARIMA periods). Third line, expectation term of Model I only: wild-bootstrap standard error of the whole three-stage chain (999 replications), marks from the centred bootstrap p-value.
[d] n.i. = not identified (no variation of the exchange rate under the Period 1 peg when it is not deflated).
[e] Significance (legend of original Tables 4.9-4.10): † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-sided; normal for the ARIMA periods, t for OLS).
[f] Periods: 1 = 2000-01 to 2005-06; 2 = 2005-07 to 2006-05; 3 = 2006-06 to 2008-05; 4 = 2008-06 to 2011-12 (Replication Plan s.3.3).

6 Effect on the conclusions

Tables 13 (Model I) and 14 (Model II) follow the expectation coefficient through the corrections, one step at a time. Row 0 is the published value, row 1 our reproduction, and row 1b the same on raw data. Rows 3 to 6 each add one correction to the row before, with the comparisons 1 → 1b (the article's deflation), 1b → 3 (GDP), 3 → 4 (fixed-base CPI) and 4 → 6 (log differences). Rows 2 and 5 replace the conventional standard errors by bootstrap ones, and row 7 is the pooled model.

Table 13: Model I: the expectation coefficient through the corrections. Source: original Tables 4.9-4.10; own estimates.

step	P1	P2	P3	P4
0 original (parentheses read as SE)	328.303†	164.243	-200.074	-4308.484
	(188.021)	(1192.054)	(476.606)	(8324.331)
1 reproduction (deflated, naive SE)	-257.454	-722.707	462.547	386.987
	(1918.094)	(1885.907)	(1016.045)	(2575.817)
1b reproduction (raw, naive SE)	n.i.	-634.416	-3106.553*	22.620
		(1621.390)	(1244.516)	(4191.952)
2 reproduction (deflated), wild bootstrap SE	-257.454	-722.707	462.547	386.987
	(1060.967)	(2221.806)	(1039.093)	(2023.897)
2 reproduction (deflated), block bootstrap SE	-257.454	-722.707	462.547	386.987

step	P1	P2	P3	P4
	(3158.649)	(3835.865)	(1122.543)	(6063.339)
3 + discrete quarterly GDP (raw)	n.i.	-634.416	-3106.553*	22.620
		(1621.390)	(1244.516)	(4191.952)
4 + fixed-base CPI, ER/NDF undeflated	n.i.	94.719	-2871.442	1476.861
		(1152.605)	(2699.493)	(3395.443)
5 corrected, wild bootstrap SE	n.i.	94.719	-2871.442	1476.861
		(1409.991)	(1558.607)	(5443.930)
6 corrected, log differences	n.i.	0.081	-1.372	0.603
		(1.082)	(1.958)	(1.101)
7 pooled full, Newey-West (deflated)	-345.232	-722.707	-1170.851	386.987
	(2072.420)	(790.469)	(1176.814)	(2211.704)
7 pooled full, Newey-West (corrected)	n.i.	94.719	-2371.261**	1476.861
		(584.259)	(850.386)	(3012.285)
7 pooled exp_only, Newey-West (deflated)	-647.148	-356.205	-937.211	349.691
	(3861.889)	(2161.778)	(1186.457)	(1935.300)
7 pooled exp_only, Newey-West (corrected)	n.i.	-918.965	-2288.598**	1556.231
		(850.136)	(864.105)	(2891.659)

[a] Coefficient on the stage-3 expected change of the exchange rate (Model I) in each period (step 7: in each regime of the pooled model), standard error below in parentheses.
[b] Step 0: original, with the numbers in parentheses read as standard errors. Step 1: our reproduction (headline data version). Step 1b: the same on raw data. Steps 3-6 add one correction at a time (README_week4 s.2.1): 1b to 3 isolates the GDP correction, 3 to 4 the fixed-base CPI.
[c] Steps 2 and 5: bootstrap standard error of the whole three-stage chain (999 replications; wild: Rademacher; block: moving blocks within periods); marks from the centred bootstrap p-value.
[d] Step 6: coefficients are elasticities (log differences of FER and of the exchange-rate terms). Step 7: pooled OLS 2000-2011 with regime-specific constants and expectation terms; full = all coefficients regime-specific; exp_only = controls common; Newey-West standard errors.
[e] n.i. = not identified.
[f] Significance (legend of original Tables 4.9-4.10): † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-sided; normal for the ARIMA periods, t for OLS).
[g] Periods: 1 = 2000-01 to 2005-06; 2 = 2005-07 to 2006-05; 3 = 2006-06 to 2008-05; 4 = 2008-06 to 2011-12 (Replication Plan s.3.3).

Table 14: Model II: the NDF coefficient through the corrections. Source: original Tables 4.9-4.10; own estimates.

step	P1	P2	P3	P4
0 original (parentheses read as SE)	109.567	305.763	-163.185	-1294.446*
	(117.109)	(395.505)	(376.356)	(531.200)
1 reproduction (deflated, naive SE)	111.435	117.091	-158.101	-1282.991*
	(116.916)	(331.001)	(318.028)	(523.642)
1b reproduction (raw, naive SE)	21.164	-182.012	-451.932	-2025.241**
	(123.751)	(577.136)	(608.588)	(595.718)
3 + discrete quarterly GDP (raw)	23.508	-181.290	-406.090	-1942.208**
	(115.146)	(526.621)	(409.807)	(600.273)
4 + fixed-base CPI, ER/NDF undeflated	8.040	-18.652	-377.679	-1506.262**
	(119.602)	(426.227)	(309.467)	(482.755)
6 corrected, log differences	-0.102	-0.057	-0.213	-0.460**
	(0.260)	(0.383)	(0.154)	(0.162)
7 pooled full, Newey-West (deflated)	121.166	117.091	162.820	-1282.991
	(156.819)	(103.866)	(314.859)	(853.688)
7 pooled full, Newey-West (corrected)	-108.413	-18.652	130.540	-1506.262*
	(222.184)	(160.708)	(344.420)	(741.241)
7 pooled exp_only, Newey-West (deflated)	126.516	441.791*	256.858	-1258.068
	(153.199)	(200.742)	(304.864)	(813.812)
7 pooled exp_only, Newey-West (corrected)	-116.098	551.521	72.686	-1609.144*
	(223.077)	(429.832)	(278.563)	(712.074)

[a] Coefficient on DNDF (Model II) in each period (step 7: in each regime of the pooled model), standard error below in parentheses.
[b] Step 0: original, with the numbers in parentheses read as standard errors. Step 1: our reproduction (headline data version). Step 1b: the same on raw data. Steps 3-6 add one correction at a time (README_week4 s.2.1): 1b to 3 isolates the GDP correction, 3 to 4 the fixed-base CPI.
[c] No bootstrap for Model II: its expectation regressor is observed, not generated.
[d] Step 6: coefficients are elasticities (log differences of FER and of the exchange-rate terms). Step 7: pooled OLS 2000-2011 with regime-specific constants and expectation terms; full = all coefficients regime-specific; exp_only = controls common; Newey-West standard errors.
[e] n.i. = not identified.
[f] Significance (legend of original Tables 4.9-4.10): † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-sided; normal for the ARIMA periods, t for OLS).
[g] Periods: 1 = 2000-01 to 2005-06; 2 = 2005-07 to 2006-05; 3 = 2006-06 to 2008-05; 4 = 2008-06 to

2011-12 (Replication Plan s.3.3).

For reference, the Model II coefficient of Period 4 is -1,294.45* (SE 531.20) as published, -1,282.99* (SE 523.64) in our reproduction, -1,506.26** (SE 482.76) on the corrected data, and -0.460** (SE 0.162) as an elasticity. For Model I, Period 3, it is -200.07 (SE 476.61) as published, 462.55 (SE 1,016.05) reproduced, and -2,871.44 (SE 2,699.49) corrected, with a wild-bootstrap standard error of 1,558.61 (p = 0.318).

[YOUR READING, period by period: at which step, if any, the sign or the significance of the expectation coefficient changes; which of the article's statements about expectations survive all corrections; which depend on one particular step; and which cannot be assessed (Period 1). Use the JCRE wording for interpretations, "Our interpretation suggests …".]

7 Reproducibility and data availability

Every result in this note is produced by the code in the replication package, **[REPOSITORY URL]**, at commit 2447209, with R version 4.5.3 (2026-03-11 ucrt) and the package versions recorded in `renv.lock`; the tables and figures were generated on 2026-10-06. The code starts from the raw files and runs in a fixed order; each script writes a run log. The bootstrap uses a fixed seed, so reruns give identical draws. **[CONFIRM from two 08_corrections logs: a rerun reproduced every estimate and bootstrap standard error.]**[5]

Public data (SAFE, IMF, FRED, NBS) are included in the package with their download dates and checksums (Table 3). Two inputs cannot be redistributed. The original document is unpublished, and its reproduction targets (the numbers of its tables) are **[included / withheld; W4-08]**. The Wind NDF series is proprietary: a reader with Wind access can export it using the code in Table 3 and run the package unchanged. Without it, everything except Model II and the NDF unit-root test can be rerun. **[CONFIRM OR EDIT]**

8 Conclusion

[YOUR READING, one paragraph each, in JCRE's four-part logic: (1) what was reproduced and what was not (Section 4); (2) which errors were found and corrected (Section 5); (3) which of the article's conclusions are confirmed and which are not (Section 6); (4) what remains open, for example the Period 1 and Period 3 constants and the Chow statistics.]

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[5] Python collects and registers the data; R estimates. Each script writes a run log to `outputs/logs/`.

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Appendix A. Decisions cited

Table 15: Decisions in the author's decision log. Each entry gives the options considered, the evidence and the date.

Decision	Title
W1-01	Location of the extracted appendix
W1-02	The original document (.docx) is not committed
W1-03	Wind NDF data
W1-04	Image-label map
W1-05	Provenance of the original document (.docx)
W1-06	Software environment
W1-07	Classification of reproduction status
W1-08	Weekly READMEs are not committed
W1-09	The decisions log is not committed
W1-10	The week-summary script is not committed
W2-01	BOP selection rule: combining DF-GLS and DW
W2-02	DF-GLS specification: calibrate on the original-study variables, then fix
W2-03	Data version for the stage-1 DW in the BOP rule
W2-04	Quarterly-to-monthly BOP conversions
W2-05	Differencing and period samples
W2-06	NDF: source file, checks and monthly versions
W2-07	Verification tolerance
W2-08	Where the analysis panel lives
W2-09	Result: DF-GLS specification selected
W2-10	Result: NDF version and BOP candidate selected
W2-11	SAFE BPM6 vintage used
W2-12	BOP series: current account only
W2-13	Tables 3.2-3.9 verification
W3-01	Differencing convention (completes W2-05)
W3-02	Rule naming the Model I specification that reproduces Tables 4.9-4.10

Decision	Title
W3-03	ARIMA(1,1,0) estimation and standard errors
W3-04	Stage 2 without a constant
W3-05	Chow tests: as specified, plus diagnostic variants
W3-06	Breusch-Godfrey specification
W3-07	Result: Model I specification reproducing Tables 4.9-4.10
W3-08	Result: Model II classification
W3-09	Result: DW, BG and Chow against Tables 4.3-4.8 and 4.11
W3-10	Raw Period 1: expectation coefficient not identified
W3-11	Sample sizes differ from the original N; within convention closer
W3-12	DW and BG statistics not reproduced to print
W3-13	Chow F statistics: degrees of freedom partly matched, magnitudes not
W3-14	Reproduced counts rest largely on insignificant cells
W3-15	Week 3 code and outputs committed
W4-01	Bootstrap for the generated regressors
W4-02	Pooled model with regime interactions
W4-03	Newey-West settings
W4-04	Scope of the corrections
W4-05	Data corrections: GDP, CPI, logs
W4-06	Selection rules for the note tables
W4-07	History check stricter than the Blueprint
W4-08	Publishing original_targets.csv; the decisions log
W4-09	Licence and rendered files
W4-13	renv sandbox switched off
W4-14	NBS published year-on-year CPI as a check of the original pi
W4-15	W3-14 settled: subsets reported in the note